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Exp-4

Q: We define F(N, y) as number of different corresponding bits in Binary representation of x and y.

eg: N=2
A = {2, 4}
4

N=3
A = {1, 3, 5}
8

Approach:
1) Loop through 31 bits
2) for each bit:
count c1 = number with set bit = 1
count c2 = number ~~of~~ with unset bits = 0
3) pairs = $c_1 * c_2$
contribution = 2^k pairs
ans += contribution
4) Return ans % MOD

Complexity: $O(31 \times N) \approx O(N)$

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Code:

```
#define ll long long
class Solution {
  public:
    int countBits(int n, vector<ll> & arr) {
      const ll mod = 1e7;
      ll totalDiff = 0;

      for (int pos = 0; pos < 5); pos++) {
          ll ones = 0;
          for (int j = 0; j < n; j++) {
              if ((arr[j] >> pos) & 1) {
                  ones++;
              }
          }

          ll zeros = n - ones;
          ll contribution = (2LL * ones * zeros) % mod;
          totalDiff = (totalDiff + contribution) % mod;
      }
      return (int) totalDiff;
    }
};
int main()
    int n; cin >> n;
    vector<ll> a(n);
    for (int i = 0; i < n; i++) cin >> a[i]
    Solution obj;
    cout << obj.countBits(n, a);
}
```